

AXECC: Benchmarking the Privacy and Accessibility Impact of Browser Extensions Supplementary Material

James M. Clarke, Ehsan Toreini, Maryam Mehrnezhad

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1 Methodology

Table 1 shows the accessibility results for all of our HoneyPages.

Table 1: Accessibility Results for our HoneyPages

HoneyPage	Lighthouse	WAVE					
		Errors	Contrast Errors	Alerts	Features	Structural Elements	ARIA
Baseline	86%	1	3	0	3	6	4
WordPress Index	96%	0	0	2	3	14	14
WordPress Login	100%	0	0	0	7	2	4
WordPress Blog	96%	0	0	2	9	20	15
WordPress 404	96%	0	0	1	6	13	13

Figure 1 shows the number of trackers recorded when using the Puppeteer Stealth plugin compared to the number of trackers recorded without it. This highlights the limited difference between the two methods, and highlights why the trade-offs with stability were not worth the benefits of using the Puppeteer Stealth plugin.

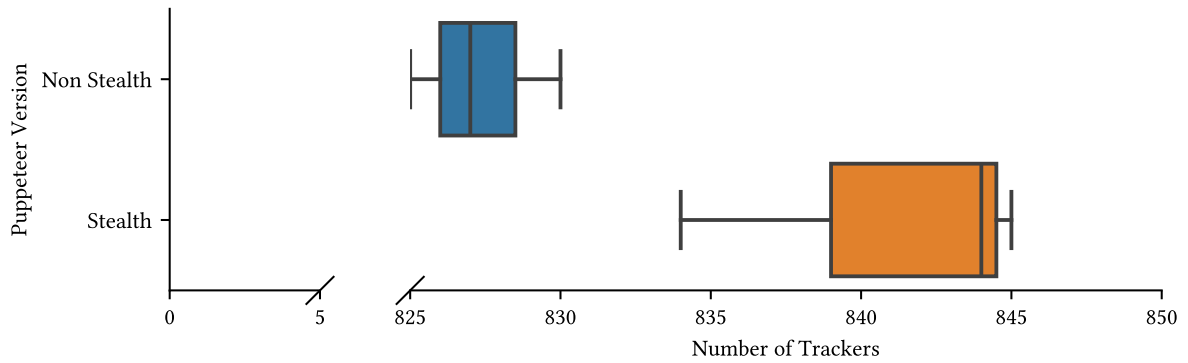


Figure 1: Box plots of the Number of Trackers for Puppeteer and Puppeteer Stealth.

Figure 2 shows the results of the crawl time experiments. As highlighted in the full paper, the number of trackers for crawl lengths from 10 to 60 seconds. It then levels out after 60 seconds, leading to the conclusion that 60 seconds is a good crawl time for our experiments.

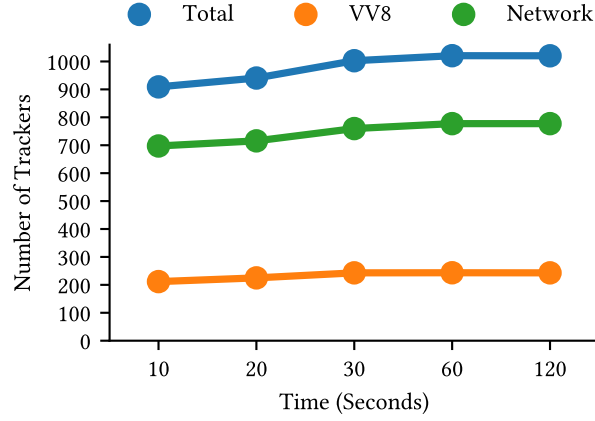


Figure 2: Results of Crawl Time Experiments

2 Static Analysis Results

Figure 3 shows the number of extensions per category which use Manifest V2, V3, or have no manifest.

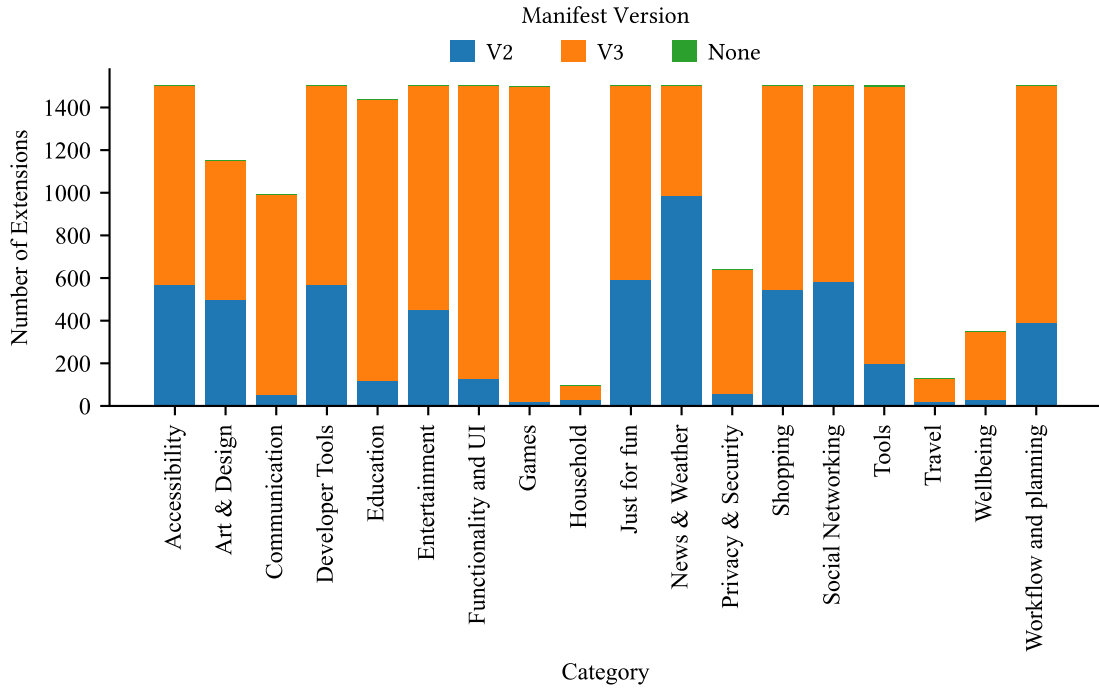


Figure 3: Manifest Version of Extensions per Category

Figure 4 shows the average number of permissions per extension per category.

Table 2: Overview of our Web Tracking Results Across Honeypages

	Extensions	Permissions					Manifest		API Calls					Ratings					No. Users				
		Mean	Median	Mode	Min	Max	V2	V3	Mean	Median	Mode	Min	Max	Mean	Median	Mode	Min	Max	Mean	Median	Mode	Min	Max
WordPress Index																							
Overall	100%	1.98	1	0	0	18	27.31%	72.61%	510	49	38	0	440K	4.04	4.6	5	0	5	57K	286	1000	0	206M
Tracking	15.63%	2.42	2	0	0	18	16.97%	83%	2316	1576	0	0	190K	4.06	4.6	5	0	5	166K	1000	1000	0	206M
Non Tracking	84.37%	1.9	1	0	0	17	29.22%	70.69%	175	44	38	0	440K	4.04	4.6	5	0	5	37K	204	1000	0	64M
WordPress Login																							
Overall	100%	1.98	1	0	0	18	27.31%	72.61%	492	39	28	0	323K	4.04	4.6	5	0	5	57K	286	1000	0	206M
Tracking	15.59%	2.42	2	0	0	18	17.01%	82.96%	2277	1551	0	0	268K	4.07	4.6	5	0	5	167K	1000	1000	0	206M
Non Tracking	84.41%	1.9	1	0	0	17	29.21%	70.7%	162	34	28	0	323K	4.04	4.6	5	0	5	37K	204	1000	0	64M
WordPress Blog																							
Overall	100%	1.98	1	0	0	18	27.31%	72.62%	498	49	38	0	485K	4.04	4.6	5	0	5	57K	286	1000	0	206M
Tracking	15.66%	2.41	2	0	0	18	17.08%	82.89%	2184	1572	0	0	220K	4.07	4.6	5	0	5	166K	1000	1000	0	206M
Non Tracking	84.34%	1.9	1	0	0	17	29.2%	70.71%	184	44	38	0	485K	4.04	4.6	5	0	5	37K	203	1000	0	64M
WordPress 404																							
Overall	100%	1.98	1	0	0	18	27.31%	72.61%	503	49	38	0	443K	4.04	4.6	5	0	5	57K	286	1000	0	206M
Tracking	15.61%	2.42	2	0	0	18	16.89%	83.08%	2205	1575	0	0	283K	4.07	4.6	5	0	5	179K	1000	1000	0	206M
Non Tracking	84.39%	1.9	1	0	0	17	29.23%	70.68%	188	44	38	0	443K	4.04	4.6	5	0	5	35K	203	1000	0	64M

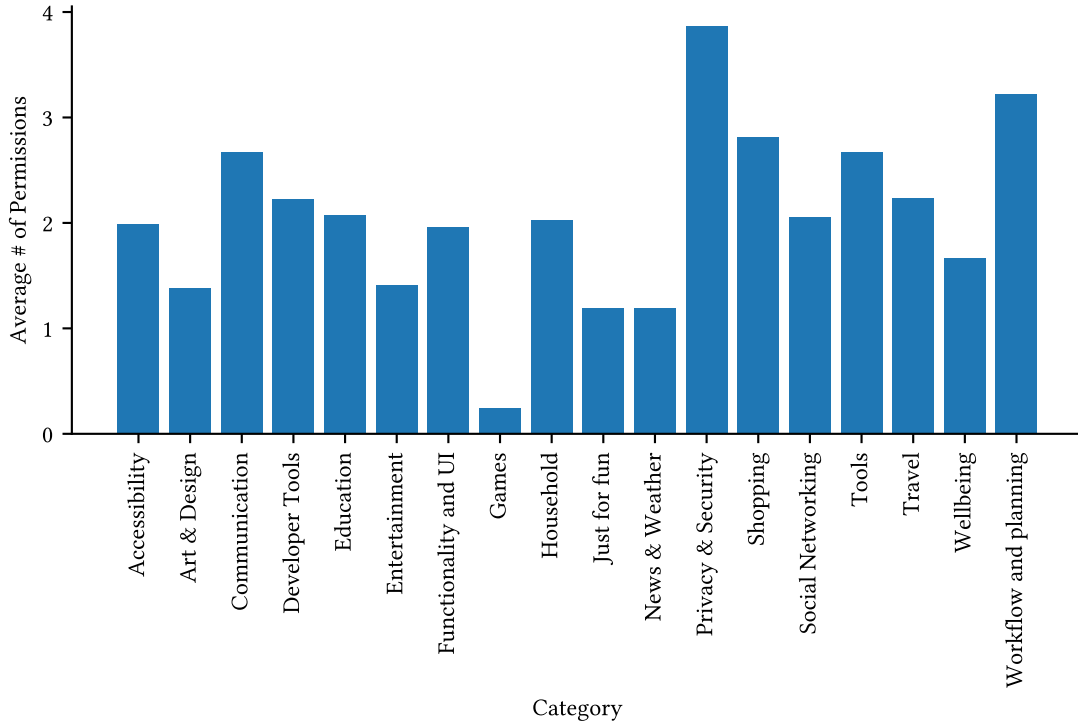


Figure 4: Average Number of Permissions of Extensions per Category

3 Web Tracking Results

Figure 5 shows the number of tracking extensions per category over all of our HoneyPages. In addition, Figure 6 shows the number of trackers per category over all of our HoneyPages.

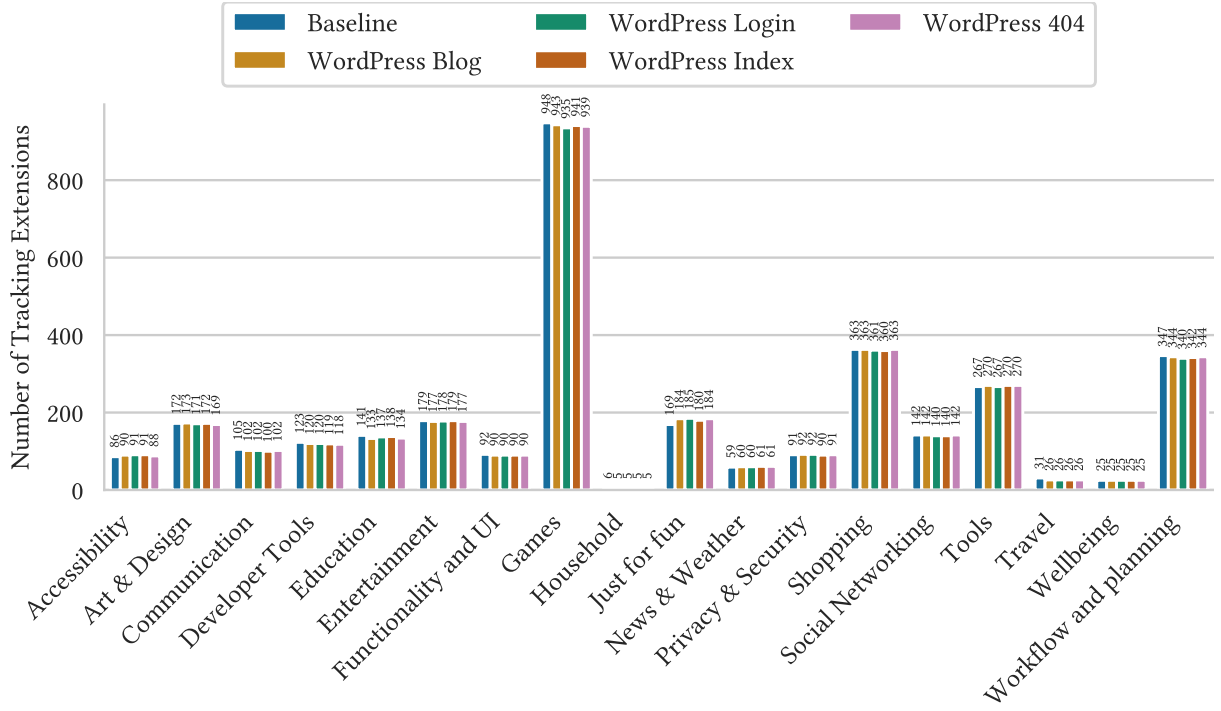


Figure 5: Number of Tracking Extensions per Category over HoneyPages

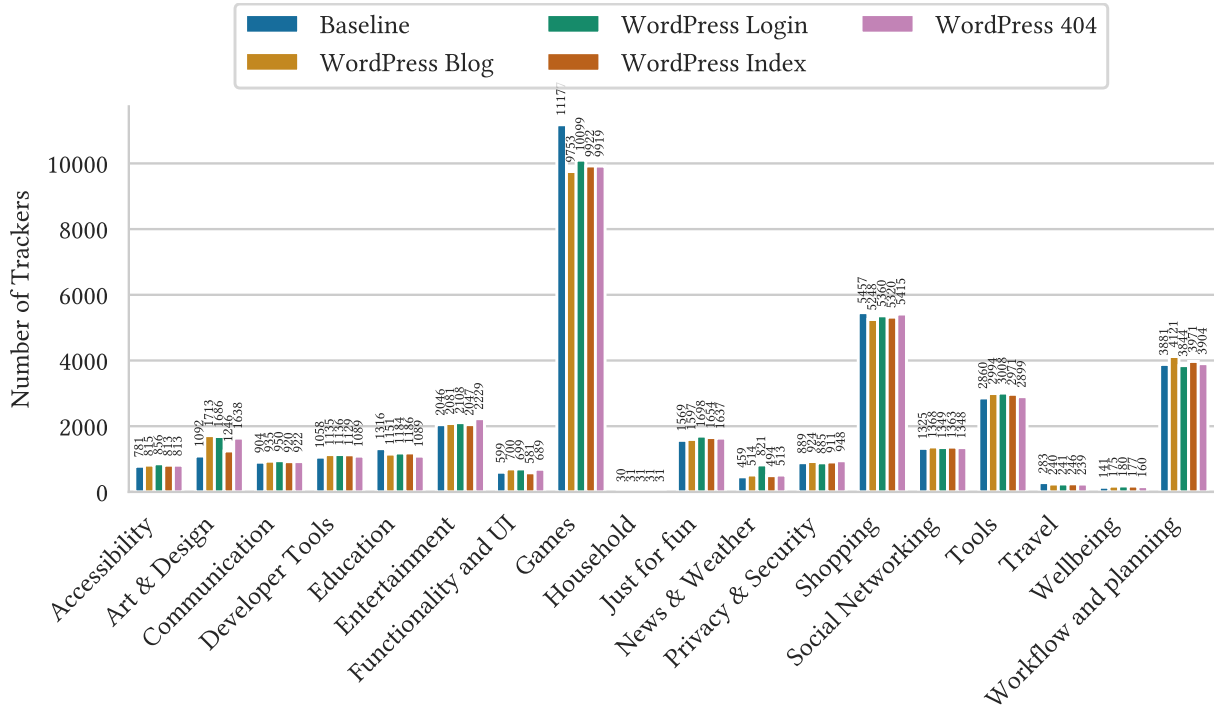


Figure 6: Number of Trackers per Category over HoneyPages

Figure 7, Figure 8, and Figure 9, and Figure 10 show the mapping of trackers from extension categories to domains to domain owners for each of our WordPress HoneyPages.

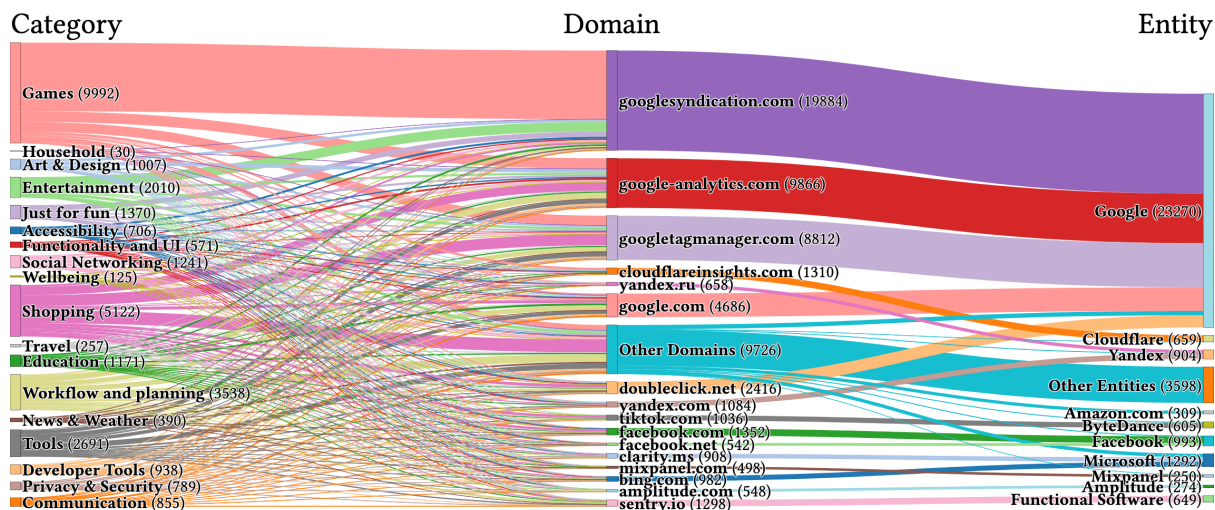


Figure 7: Mapping of Trackers, from Extension Categories to Domains to Domain Owner for the WordPress Index HoneyPage. (Other is less than 400.)

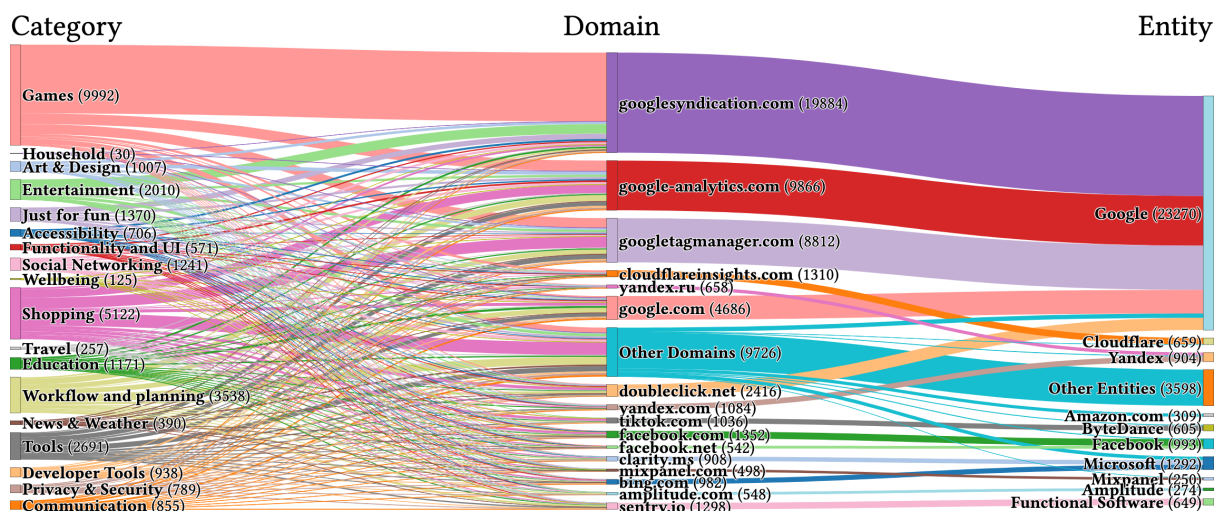


Figure 8: Mapping of Trackers, from Extension Categories to Domains to Domain Owner for the WordPress Login HoneyPage. (Other is less than 400.)

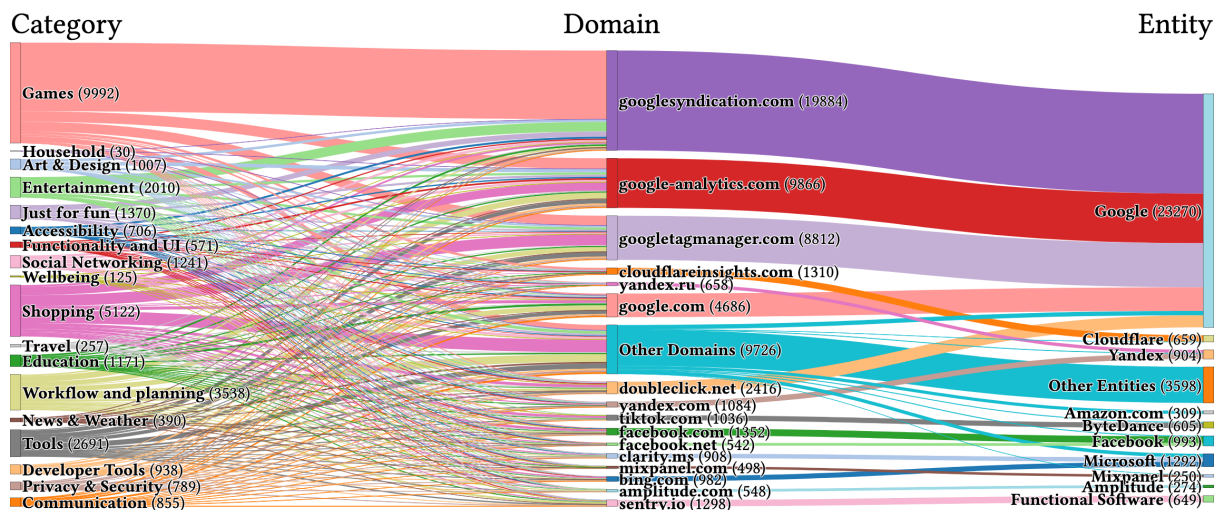


Figure 9: Mapping of Trackers, from Extension Categories to Domains to Domain Owner for the WordPress Blog HoneyPage. (Other is less than 400.)

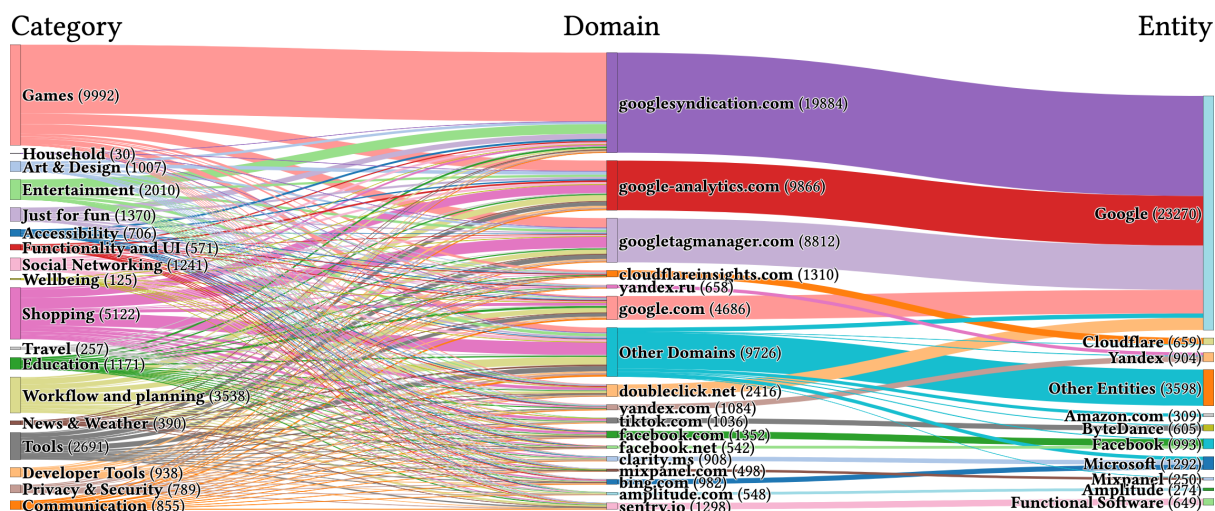


Figure 10: Mapping of Trackers, from Extension Categories to Domains to Domain Owner for the WordPress 404 HoneyPage. (Other is less than 400.)

Figure 11, Figure 12, Figure 13, and Figure 14 show the boxplots of the number of trackers for each of our WordPress HoneyPages.

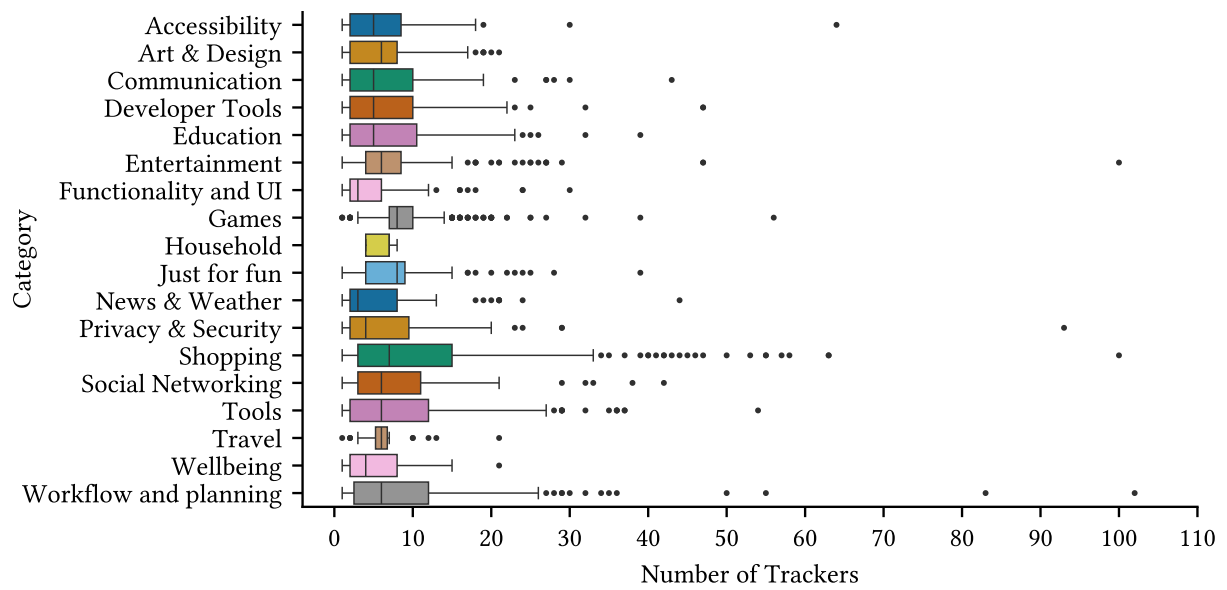


Figure 11: Boxplot of the Number of Trackers for the WordPress Index HoneyPage.

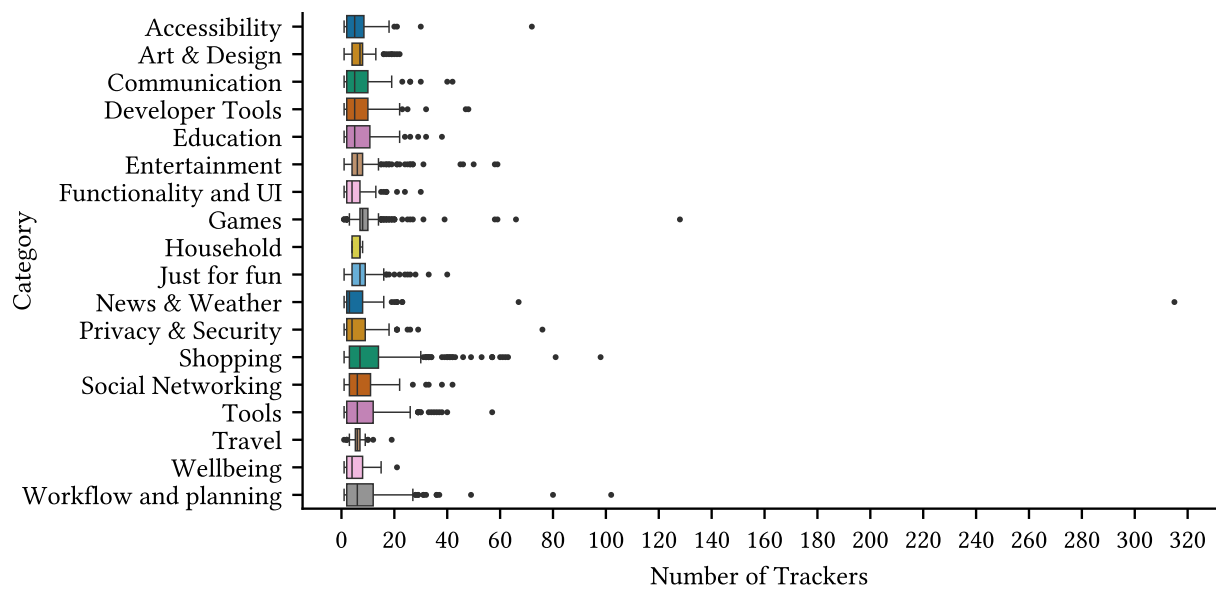


Figure 12: Boxplot of the Number of Trackers for the WordPress Login HoneyPage.

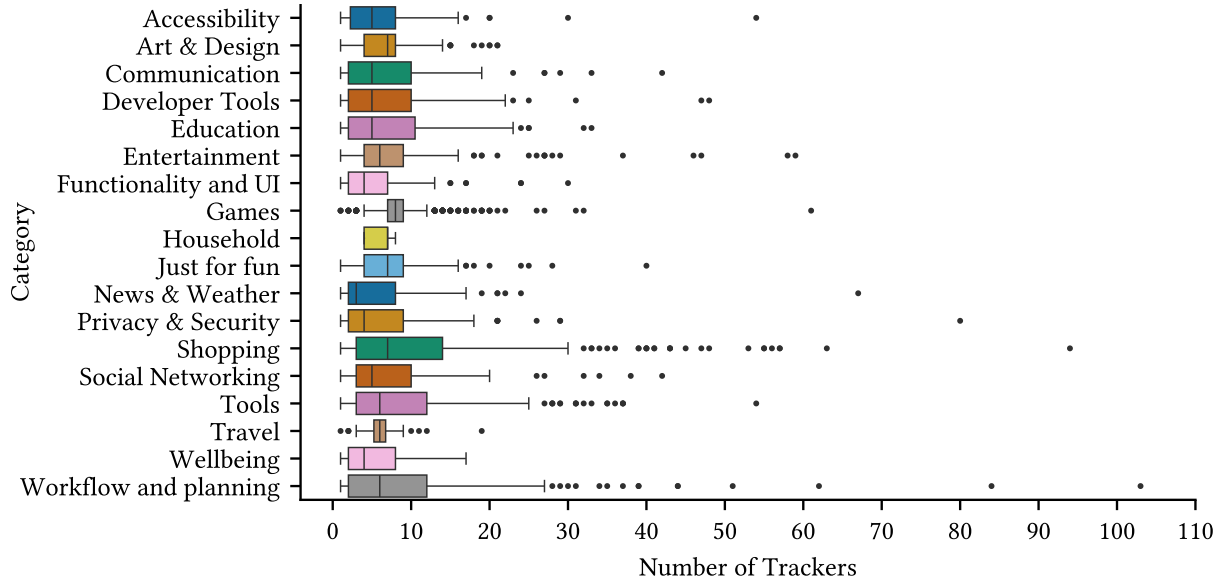


Figure 13: Boxplot of the Number of Trackers for the WordPress Blog HoneyPage.

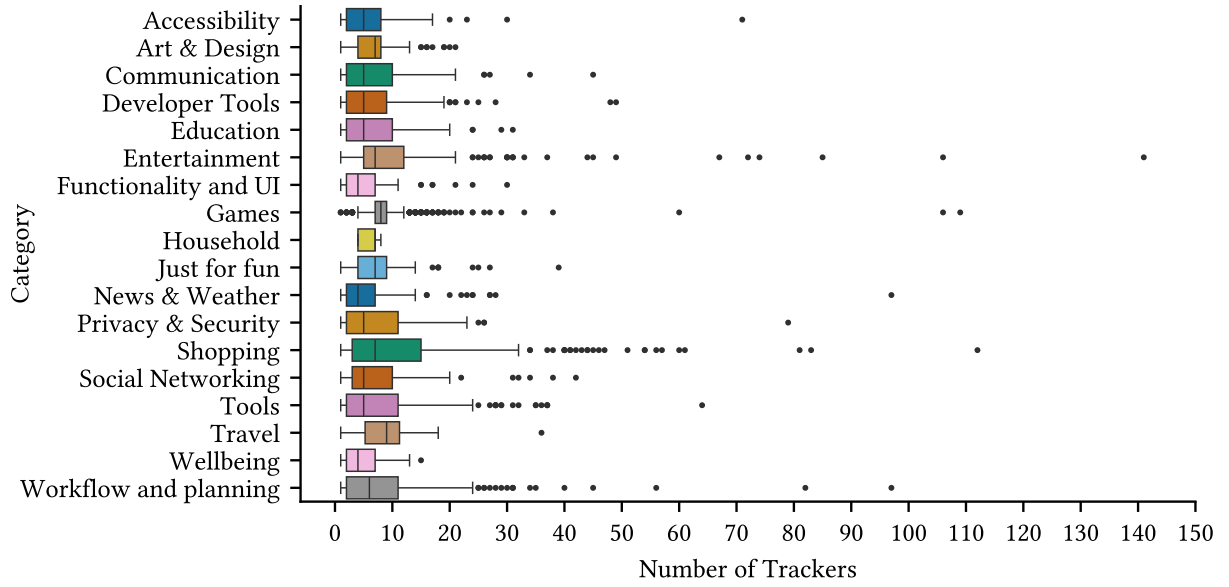


Figure 14: Boxplot of the Number of Trackers for the WordPress 404 HoneyPage.

Table 3: Summary of Tracking Detection Results

HoneyPage	Detection Mechanism		
	VV8	Network	Total
Baseline	6932	28935	35867
WordPress Index	7343	27639	34982
WordPress Login	7738	28397	36135
WordPress Blog	7835	27660	35495
WordPress 404	7839	27643	35482

Table 4: Top 10 Extensions by Number of Trackers for Each HoneyPage

(a) Baseline		(b) WordPress Index	
Extension	Trackers	Extension	Trackers
GWA Calculator Export GWA Results	65	Read Pronunciation English & French	43
AMBOSS Medical Knowledge Everywhere	42	QuestionAI Homework Powered AI Assistant	32
Read Pronunciation English & French	41	Language Learning with Disney Plus(Beta)-AFL	30
Language Project	35	Language Learning with Netflix & YouTube-AFL	29
QuestionAI Homework Powered AI Assistant	34	Video Notes to Notion & AI Flashcard by Snipo	27
Speechify Text to Speech Voice Reader	31	Interactive Dual Subs for Languages Learning	26
Interactive Dual Subs for Languages Learning	30	Grade Grabber	25
Black Menu for Wikipedia	29	Speechify Text to Speech Voice Reader	25
Ddict Translate Translator - Dictionary	29	Bytelearn- AI Math Tutor	24
Language Learning with Netflix & YouTube-AFL	29	Mouse Tooltip Translator	23

(c) WordPress Login		(d) WordPress Blog	
name	count	name	count
Read Pronunciation English & French	40	Read Pronunciation English & French	35
QuestionAI Homework Powered AI Assistant	32	QuestionAI Homework Powered AI Assistant	32
Interactive Dual Subs for Languages Learning	31	Language Learning with Disney Plus(Beta)-AFL	29
Language Learning with Disney Plus(Beta)-AFL	30	Language Learning with Netflix & YouTube-AFL	29
Language Learning with Netflix & YouTube-AFL	28	Video Notes to Notion & AI Flashcard by Snipo	27
Video Notes to Notion & AI Flashcard by Snipo	27	Interactive Dual Subs for Languages Learning	26
Bytelearn- AI Math Tutor	25	Grade Grabber	25
Grade Grabber	25	Speechify Text to Speech Voice Reader	25
Speechify Text to Speech Voice Reader	25	Bytelearn- AI Math Tutor	24
StudyX Your AI Homework Writing & Reading Assistant	23	Mouse Tooltip Translator	23

(e) WordPress 404	
name	count
QuestionAI Homework Powered AI Assistant	31
Read Pronunciation English & French	31
Interactive Dual Subs for Languages Learning	26
Speechify Text to Speech Voice Reader	25
Video Notes to Notion & AI Flashcard by Snipo	25
Bytelearn- AI Math Tutor	24
Language Learning with Disney Plus(Beta)-AFL	24
Language Learning with Netflix & YouTube-AFL	24
Mouse Tooltip Translator	24
StudyX Your AI Homework Writing & Reading Assistant	24

Table 5: Top JavaScript APIs used by Extensions on Different HoneyPages.

(a) Baseline		(b) WordPress Index		(c) WordPress Login	
API Name	Count	API Name	Count	API Name	Count
HTMLInputElement	1425076	HTMLInputElement	1641687	HTMLInputElement	1725451
HTMLDivElement	1220423	HTMLDivElement	1350621	HTMLDivElement	1194974
Object	1077093	HTMLAnchorElement	768086	Object	827489
HTMLAnchorElement	740391	HTMLDocument	759670	HTMLAnchorElement	747437
CSSStyleDeclaration	536607	Object	666041	HTMLDocument	593829
HTMLDocument	452547	CSSStyleDeclaration	519911	CSSStyleDeclaration	507853
IDBRequest	341277	Window	285528	Window	328593
Window	284973	IDBRequest	264465	IDBRequest	321539
HTMLImageElement	274723	HTMLImageElement	263378	HTMLImageElement	252210
HTMLScriptElement	211760	DOMTokenList	244776	HTMLScriptElement	213870

(d) WordPress Blog		(e) WordPress 404	
API Name	Count	API Name	Count
HTMLInputElement	1373168	HTMLInputElement	1370165
HTMLDivElement	1359472	HTMLDivElement	1317072
HTMLDocument	761686	Object	836141
HTMLAnchorElement	753057	HTMLDocument	756916
Object	700123	HTMLAnchorElement	754819
CSSStyleDeclaration	535151	CSSStyleDeclaration	517382
Window	286903	IDBRequest	309669
HTMLImageElement	258618	Window	286516
Element	254759	Element	273860
DOMTokenList	242369	HTMLImageElement	265484

Table 6: Top JavaScript APIs used by Extensions on Different HoneyPages which could be used for Tracking.

(a) Baseline		(b) WordPress Index	
API Name	Count	API Name	Count
Object.addListener	907136	HTMLInputElement.addEventListener	570591
HTMLInputElement.addEventListener	570249	HTMLInputElement.setAttribute	556560
HTMLInputElement.setAttribute	556024	Object.addListener	498119
HTMLAnchorElement.addEventListener	442055	HTMLAnchorElement.addEventListener	448229
CSSStyleDeclaration.getPropertyValue	400966	HTMLDivElement.appendChild	376010
HTMLDivElement.appendChild	335402	CSSStyleDeclaration.getPropertyValue	374201
HTMLInputElement.getAttribute	201575	HTMLDocument.createElement	299270
IDBRequest.removeEventListener	193646	HTMLInputElement.dispatchEvent	289205
HTMLDivElement.setAttribute	182776	HTMLInputElement.getAttribute	209883
HTMLDocument.createElement	178084	HTMLDivElement.setAttribute	186965
(c) WordPress Login		(d) WordPress Blog	
API Name	Count	API Name	Count
Object.addListener	658955	HTMLInputElement.addEventListener	570069
HTMLInputElement.addEventListener	569302	Object.addListener	533105
HTMLInputElement.setAttribute	555632	HTMLAnchorElement.addEventListener	441818
HTMLAnchorElement.addEventListener	440870	HTMLDivElement.appendChild	393228
CSSStyleDeclaration.getPropertyValue	366330	CSSStyleDeclaration.getPropertyValue	387650
HTMLDivElement.appendChild	307336	HTMLDocument.createElement	302266
HTMLInputElement.dispatchEvent	287744	HTMLInputElement.dispatchEvent	288940
HTMLDocument.createElement	216757	HTMLInputElement.setAttribute	288672
HTMLInputElement.getAttribute	208927	Element.getAttribute	241088
IDBRequest.removeEventListener	184288	HTMLInputElement.getAttribute	208337
(e) WordPress 404			
API Name	Count		
Object.addListener	669542		
HTMLInputElement.addEventListener	570422		
HTMLAnchorElement.addEventListener	445099		
CSSStyleDeclaration.getPropertyValue	371641		
HTMLDivElement.appendChild	370815		
HTMLDocument.createElement	301432		
HTMLInputElement.setAttribute	288825		
HTMLInputElement.dispatchEvent	287786		
Element.getAttribute	260363		
HTMLInputElement.getAttribute	207582		

Table 7: Top 10 JavaScript APIs used by Tracking Extensions and Non-Tracking Extensions

(a) Baseline

Tracking		Non-Tracking	
API Name	Percent	API Name	Percent
HTMLInputElement	18.35	Object	22.61
HTMLAnchorElement	9.54	HTMLDivElement	20.12
HTMLDivElement	9.17	IDBRequest	8.83
Object	6.40	HTMLDocument	5.03
CSSStyleDeclaration	6.36	Element	3.26
HTMLDocument	4.25	Window	2.60
HTMLImageElement	2.92	CSSStyleDeclaration	2.21
Window	2.88	HTMLImageElement	2.08
HTMLScriptElement	2.64	HTMLInputElement	1.72
HTMLButtonElement	1.94	HTMLFieldSetElement	1.66

(b) WordPress Index

Tracking		Non-Tracking	
API Name	Percent	API Name	Percent
HTMLInputElement	20.95	HTMLDivElement	20.03
HTMLAnchorElement	9.59	HTMLDocument	12.58
HTMLDivElement	9.39	Object	12.34
CSSStyleDeclaration	5.99	IDBRequest	4.99
HTMLDocument	4.74	OffscreenCanvasRenderingContext2D	3.84
Object	3.62	DOMTokenList	2.99
Window	2.86	Element	2.73
HTMLImageElement	2.83	Storage	2.48
HTMLScriptElement	2.67	Window	2.12
HTMLSpanElement	2.04	HTMLBodyElement	2.00

(c) WordPress Login

Tracking		Non-Tracking	
API Name	Percent	API Name	Percent
HTMLInputElement	21.29	HTMLDivElement	16.72
HTMLAnchorElement	9.54	Object	16.15
HTMLDivElement	9.34	HTMLDocument	8.72
CSSStyleDeclaration	5.90	IDBRequest	7.34
Object	4.70	OffscreenCanvasRenderingContext2D	4.13
HTMLDocument	4.48	HTMLInputElement	3.92
Window	2.99	Window	3.51
HTMLImageElement	2.80	Element	2.94
HTMLScriptElement	2.65	CSSStyleDeclaration	2.12
HTMLSpanElement	1.88	Global	2.04

(d) WordPress Blog

Tracking		Non-Tracking	
API Name	Percent	API Name	Percent
HTMLInputElement	18.27	HTMLDivElement	20.18
HTMLAnchorElement	9.84	HTMLDocument	11.93
HTMLDivElement	9.47	Object	11.90
CSSStyleDeclaration	6.27	Element	5.74
HTMLDocument	5.03	IDBRequest	4.05
Object	4.20	OffscreenCanvasRenderingContext2D	3.65
Window	3.03	DOMTokenList	3.03
HTMLImageElement	2.89	Storage	2.36
HTMLScriptElement	2.71	CSSStyleDeclaration	2.35
DOMTokenList	1.95	HTMLSpanElement	2.28

(e) WordPress 404

Tracking		Non-Tracking	
API Name	Percent	API Name	Percent
HTMLInputElement	18.14	HTMLDivElement	18.28
HTMLAnchorElement	9.87	Object	13.85
HTMLDivElement	9.55	HTMLDocument	11.71
CSSStyleDeclaration	6.23	IDBRequest	6.01
Object	5.03	Element	5.88
HTMLDocument	4.93	OffscreenCanvasRenderingContext2D	3.59
HTMLImageElement	3.01	DOMTokenList	2.79
Window	3.00	Storage	2.32
HTMLScriptElement	2.71	HTMLSpanElement	2.03
DOMTokenList	1.95	Window	1.96

Table 8: Top 10 JavaScript APIs used by HoneyPages without an Extension (Baseline has None).

(a) WordPress Index		(b) WordPress Login	
API Name	Percent	API Name	Percent
HTMLDocument	39.47	HTMLDocument	25.00
OffscreenCanvasRenderingContext2D	15.79	OffscreenCanvasRenderingContext2D	21.43
Storage	10.53	HTMLInputElement	14.29
DOMTokenList	7.89	Window	7.14
HTMLBodyElement	7.89	Storage	7.14
HTMLHeadElement	7.89	DOMTokenList	3.57
URL	2.63	URL	3.57
Worker	2.63	Worker	3.57
DedicatedWorkerGlobalScope	2.63	HTMLBodyElement	3.57
OffscreenCanvas	2.63	HTMLHeadElement	3.57

(c) WordPress Blog		(d) WordPress 404	
API Name	Percent	API Name	Percent
HTMLDocument	20.26	HTMLDocument	39.47
HTMLScriptElement	7.72	OffscreenCanvasRenderingContext2D	15.79
HTMLMetaElement	6.99	Storage	10.53
HTMLDivElement	6.90	HTMLHeadElement	7.89
Storage	6.23	HTMLBodyElement	7.89
HTMMLinkElement	5.42	DOMTokenList	7.89
OffscreenCanvasRenderingContext2D	5.42	URL	2.63
Window	4.42	Worker	2.63
HTMLHeadElement	3.47	DedicatedWorkerGlobalScope	2.63
DOMTokenList	3.11	OffscreenCanvas	2.63

Table 9: Top 10 JavaScript APIs used by Extensions by Script Type.

(a) For the Baseline HoneyPage.

No Scripts		Content Only		Background Only		Both	
API Name	Count	API Name	Count	API Name	Count	API Name	Count
Object	123	HTMLDivElement	8611	HTMLInputElement	1331475	HTMLDivElement	938290
HTMLDivElement	96	HTMLDocument	7526	Object	902585	IDBRequest	331814
HTMLDocument	88	CSSStyleDeclaration	3837	HTMLAnchorElement	520912	HTMLDocument	284813
Window	38	HTMLInputElement	2967	CSSStyleDeclaration	287727	CSSStyleDeclaration	245043
HTMLBodyElement	28	HTMLFieldSetElement	2588	HTMLDivElement	273426	HTMLAnchorElement	218950
SVGSVGElement	27	Window	1819	HTMLImageElement	194056	Object	172929
HTMLButtonElement	24	DOMTokenList	1688	HTMLDocument	160120	Window	152947
SVGPathElement	24	Object	1456	Element	143032	HTMLSpanElement	138705
HTMLAnchorElement	24	HTMLElement	1301	Window	130169	HTMLScriptElement	127904
ServiceWorkerGlobalScope	23	HTMLHtmlElement	1076	HTMLScriptElement	83816	DOMTokenList	120403

(b) WordPress Index HoneyPage.

No Scripts		Content Only		Background Only		Both	
API Name	Count	API Name	Count	API Name	Count	API Name	Count
HTMLDocument	40307	HTMLDocument	49883	HTMLInputElement	1554324	HTMLDivElement	1057106
OffscreenCanvasRenderingContext2D	16116	OffscreenCanvasRenderingContext2D	16952	HTMLAnchorElement	530772	HTMLDocument	406473
Storage	10744	Storage	11440	Object	495099	IDBRequest	254757
HTMLBodyElement	8060	DOMTokenList	11323	CSSStyleDeclaration	301581	HTMLAnchorElement	235385
HTMLHeadElement	8058	HTMLDivElement	10459	HTMLDivElement	283044	CSSStyleDeclaration	212684
DOMTokenList	8058	HTMLBodyElement	8989	HTMLDocument	263007	HTMLSpanElement	175119
OffscreenCanvas	2686	HTMLHeadElement	8694	HTMLImageElement	177086	Object	169477
URL	2686	CSSStyleDeclaration	5646	Element	138209	Window	154708
Worker	2686	DedicatedWorkerGlobalScope	2973	Window	128970	DOMTokenList	147435
DedicatedWorkerGlobalScope	2686	Worker	2882	HTMLScriptElement	84502	HTMLScriptElement	130569

(c) WordPress Login HoneyPage.

No Scripts		Content Only		Background Only		Both	
API Name	Count	API Name	Count	API Name	Count	API Name	Count
HTMLDocument	18791	HTMLDocument	27229	HTMLInputElement	1574195	HTMLDivElement	906387
OffscreenCanvasRenderingContext2D	16080	OffscreenCanvasRenderingContext2D	16860	Object	655388	HTMLDocument	335203
HTMLInputElement	10749	HTMLInputElement	13962	HTMLAnchorElement	522113	IDBRequest	311932
Window	5381	HTMLDivElement	8385	CSSStyleDeclaration	297312	HTMLAnchorElement	224439
Storage	5360	Window	7485	HTMLDivElement	280190	CSSStyleDeclaration	206463
HTMLBodyElement	2682	Storage	5766	HTMLDocument	212606	Window	172004
URL	2680	DOMTokenList	4923	HTMLImageElement	176254	Object	170641
DOMTokenList	2680	CSSStyleDeclaration	4078	Window	143723	HTMLSpanElement	136189
Worker	2680	HTMLBodyElement	3340	Element	121304	HTMLScriptElement	130462
DedicatedWorkerGlobalScope	2680	HTMLHeadElement	3030	HTMLScriptElement	82812	HTMLInputElement	126545

(d) WordPress Blog HoneyPage.

No Scripts		Content Only		Background Only		Both	
API Name	Count	API Name	Count	API Name	Count	API Name	Count
HTMLDocument	40322	HTMLDocument	49884	HTMLInputElement	1283030	HTMLDivElement	1082172
OffscreenCanvasRenderingContext2D	16122	OffscreenCanvasRenderingContext2D	16950	Object	528745	HTMLDocument	402794
Storage	10748	DOMTokenList	12558	HTMLAnchorElement	520693	IDBRequest	231343
HTMLBodyElement	8063	HTMLDivElement	11866	CSSStyleDeclaration	299265	HTMLAnchorElement	230021
HTMLHeadElement	8061	Storage	11437	HTMLDocument	268686	CSSStyleDeclaration	228841
DOMTokenList	8061	HTMLBodyElement	8988	HTMLDivElement	265422	HTMLSpanElement	172300
URL	2687	HTMLHeadElement	8694	Element	238884	Object	169897
Worker	2687	CSSStyleDeclaration	7045	HTMLImageElement	175828	Window	153576
DedicatedWorkerGlobalScope	2687	DedicatedWorkerGlobalScope	2973	Window	131475	DOMTokenList	151762
OffscreenCanvas	2687	Worker	2882	HTMLScriptElement	82688	HTMLScriptElement	127402

(e) WordPress 404 HoneyPage.

No Scripts		Content Only		Background Only		Both	
API Name	Count	API Name	Count	API Name	Count	API Name	Count
HTMLDocument	40322	HTMLDocument	49863	HTMLInputElement	1283478	HTMLDivElement	1034266
OffscreenCanvasRenderingContext2D	16122	OffscreenCanvasRenderingContext2D	16950	Object	665355	HTMLDocument	397593
Storage	10748	Storage	11439	HTMLAnchorElement	525270	IDBRequest	299982
HTMLBodyElement	8063	DOMTokenList	11283	CSSStyleDeclaration	300282	HTMLAnchorElement	227913
HTMLHeadElement	8061	HTMLDivElement	10237	HTMLDivElement	272557	CSSStyleDeclaration	211463
DOMTokenList	8061	HTMLBodyElement	8967	HTMLDocument	269138	Object	169318
DedicatedWorkerGlobalScope	2687	HTMLHeadElement	8694	Element	258064	HTMLSpanElement	167441
URL	2687	CSSStyleDeclaration	5637	HTMLImageElement	184496	Window	152725
Worker	2687	DedicatedWorkerGlobalScope	2973	Window	131954	DOMTokenList	149026
OffscreenCanvas	2687	Worker	2882	HTMLScriptElement	83175	HTMLScriptElement	127443

Table 10: Number of Extensions with Content Scripts, Background Scripts, or Both.

Category	No Scripts		Content Only		Background Only		Both	
	Count	Percent	Count	Percent	Count	Percent	Count	Percent
Accessibility	122	8.1	327	21.7	413	27.5	642	42.7
Art & Design	269	23.4	115	10.0	477	41.4	291	25.3
Communication	62	6.2	103	10.4	161	16.2	666	67.1
Developer Tools	247	16.4	148	9.8	518	34.4	591	39.3
Education	196	13.6	219	15.2	327	22.7	696	48.4
Entertainment	232	15.4	335	22.3	377	25.1	559	37.2
Just for fun	334	22.2	244	16.2	480	31.9	446	29.7
Functionality and UI	159	10.6	390	25.9	457	30.4	498	33.1
Games	86	5.7	29	1.9	1314	87.8	68	4.5
Household	18	18.8	9	9.4	34	35.4	35	36.5
News & Weather	385	25.6	239	15.9	575	38.2	305	20.3
Privacy & Security	60	9.4	39	6.1	209	32.7	331	51.8
Shopping	64	4.3	131	8.7	205	13.6	1102	73.4
Social Networking	145	9.7	281	18.7	350	23.3	726	48.3
Tools	144	9.6	109	7.3	465	30.9	785	52.2
Travel	18	14.1	13	10.2	26	20.3	71	55.5
Wellbeing	74	21.0	42	11.9	115	32.7	121	34.4
Workflow and planning	116	7.7	78	5.2	436	29.0	874	58.1

Table 15 shows the average number of permissions by script type across categories. We see that extensions with both content and background scripts have the highest average number of permissions across all categories except for Accessibility, Developer Tools, and Functionality and UI.

4 Comparison of Tracking and Accessibility across Extensions Categories

Table 16 shows a summary of the comparison of tracking and accessibility across extensions categories across all of our HoneyPages.

Table 11: Average Number of Trackers by Script Type Across HoneyPages.

HoneyPage	No Scripts	Content Only	Background Only	Both
Baseline	0.00	0.01	2.65	1.98
WordPress Index	0.00	0.02	2.46	2.03
WordPress Login	0.00	0.02	2.56	2.08
WordPress Blog	0.00	0.02	2.54	2.02
WordPress 404	0.00	0.02	2.56	2.00

Table 12: Average Number of API Calls by Script Type Across HoneyPages.

HoneyPage	No Scripts	Content Only	Background Only	Both
Baseline	0	17	760	549
WordPress Index	37	58	764	595
WordPress Login	28	45	769	561
WordPress Blog	37	61	740	590
WordPress 404	37	58	766	582

Table 13: Permissions by Mean Number of Trackers.

Permission	Baseline	WordPress Pages				Overall
		WordPress Login	WordPress 404	WordPress Blog	WordPress Index	
search	18.25	22.25	21.75	22.00	23.50	21.55
system.display	17.61	17.83	17.64	17.89	17.56	17.71
idle	16.85	18.19	16.85	17.31	16.62	17.16
tts	15.42	17.45	17.36	17.55	15.00	16.56
desktopCapture	14.20	16.05	16.00	15.95	15.65	15.57
power	15.50	15.50	15.50	15.50	15.50	15.50
cookies	14.70	14.89	14.86	15.07	14.80	14.86
sessions	14.50	14.75	14.75	14.75	14.75	14.70
declarativeNetRequestWithHostAccess	13.59	15.32	14.37	14.85	14.15	14.45
webNavigation	12.70	13.08	13.26	12.91	13.36	13.06
clipboardRead	12.62	13.03	12.95	12.95	13.00	12.91
webRequest	12.85	12.77	12.62	12.67	12.64	12.71
background	12.77	12.50	12.65	13.12	12.44	12.70
debugger	11.87	12.07	12.27	12.36	12.71	12.25
identity.email	11.23	11.74	12.41	12.32	12.00	11.94
management	11.41	11.64	11.64	11.97	12.25	11.78
tabs	11.50	11.72	11.83	11.70	11.65	11.68
fontSettings	10.67	11.83	11.83	12.17	11.83	11.67
unlimitedStorage	11.41	11.05	11.86	11.88	11.83	11.61
tabCapture	10.27	11.93	11.70	11.96	12.04	11.58
scripting	11.23	11.54	11.39	11.54	11.41	11.42
sidePanel	11.26	11.08	11.83	11.12	11.41	11.34
notifications	11.17	11.34	11.23	11.58	11.33	11.33
alarms	11.36	11.28	11.25	11.34	11.04	11.26
offscreen	10.05	11.39	11.41	11.55	11.47	11.17
browsingData	10.50	11.07	11.11	11.04	11.46	11.04
None	11.55	10.60	11.32	10.73	10.98	11.04
clipboardWrite	11.21	10.99	10.88	11.22	10.88	11.03
history	10.92	10.13	10.97	10.92	10.95	10.78
contextMenus	10.49	10.85	11.05	10.82	10.52	10.74
identity	10.55	10.50	10.76	10.74	10.36	10.58
storage	10.26	10.48	10.65	10.65	10.50	10.51
activeTab	10.34	10.62	10.48	10.57	10.50	10.50
declarativeContent	9.90	10.64	10.68	9.27	11.18	10.34
bookmarks	10.87	9.89	10.36	9.86	10.04	10.20
windows	8.89	10.67	10.89	10.22	10.22	10.18
gcm	10.89	9.88	10.04	10.04	9.96	10.16
proxy	9.79	10.20	9.63	10.33	10.82	10.15
commands	9.21	10.31	10.69	10.31	9.31	9.97
topSites	8.09	7.58	10.27	9.84	10.02	9.16
declarativeNetRequest	9.07	9.07	9.11	9.43	9.10	9.16
webRequestBlocking	9.13	9.12	8.96	9.13	9.05	9.08
webRequestAuthProvider	8.27	8.09	8.36	8.77	9.95	8.69
downloads	8.72	8.67	8.52	8.69	8.44	8.61
nativeMessaging	8.79	8.50	8.50	8.29	8.21	8.46
geolocation	7.60	8.63	8.21	8.20	8.25	8.18
contentSettings	7.43	7.57	7.71	7.71	7.57	7.60
declarativeNetRequestFeedback	7.03	7.42	7.47	7.52	7.52	7.39
system.cpu	6.00	7.67	7.33	7.67	7.67	7.27
system.memory	6.14	7.00	7.00	7.17	7.17	6.90
privacy	7.05	6.65	6.91	6.70	7.05	6.87
tabGroups	6.00	6.38	6.58	6.31	6.23	6.30
chrome://favicon/	5.31	6.38	5.85	6.23	5.31	5.82
favicon	6.07	5.62	5.62	5.62	5.31	5.65
system.storage	3.50	6.00	6.00	6.00	6.00	5.50
downloads.open	5.50	5.50	4.50	5.50	4.50	5.10
fileBrowserHandler	4.50	4.50	4.50	4.50	4.50	4.50
pageCapture	4.33	4.33	4.33	4.33	4.33	4.33
downloads.shelf	4.00	4.00	3.33	4.00	3.33	3.73

Table 14: Average Number of JS API Calls by Script Type for the Baseline HoneyPage.

Category	No Scripts	Content Only	Background Only	Both
Accessibility	0	14	93	424
Art & Design	0	29	369	338
Communication	0	20	150	545
Developer Tools	0	68	222	256
Education	0	25	147	751
Entertainment	0	2	342	279
Just for fun	0	20	421	383
Functionality and UI	0	6	203	303
Games	0	1	2292	426
Household	0	14	132	143
News & Weather	0	19	134	1359
Privacy & Security	0	29	344	435
Shopping	0	10	404	767
Social Networking	0	5	2578	392
Tools	0	12	277	633
Travel	0	15	68	441
Wellbeing	0	18	48	439
Workflow and planning	4	54	370	739
Overall	0	17	760	549

Table 15: Average Number of Permissions by Script Type (Bold highlights the highest average).

Category	No Scripts	Content Only	Background Only	Both
Accessibility	1.55	1.03	2.65	2.56
Art & Design	1.01	1.01	1.75	2.58
Communication	1.11	1.17	2.74	3.17
Developer Tools	1.38	1.30	2.76	2.59
Education	1.10	1.11	2.45	2.87
Entertainment	1.04	0.99	1.88	2.38
Just for fun	1.02	0.95	1.65	2.10
Functionality and UI	1.65	1.03	2.77	2.75
Games	0.90	1.00	1.02	2.54
Household	1.17	0.89	2.00	3.26
News & Weather	0.95	0.93	1.74	2.12
Privacy & Security	1.27	1.28	3.91	4.71
Shopping	1.55	0.99	2.96	3.17
Social Networking	1.28	1.02	2.38	2.76
Tools	1.32	1.13	2.90	3.16
Travel	0.78	1.08	1.50	3.46
Wellbeing	1.01	1.17	2.20	2.40
Workflow and planning	1.58	1.36	3.42	3.61
Overall	1.18	1.05	2.18	2.93

Table 16: Comparison of Tracking and Accessibility across Extensions Categories Across HoneyPages.

(a) Tracking and API Calls

Category	% of Total Trackers					% of Extensions that Track					Mean API Calls				
	404	Baseline	Blog	Index	Login	404	Baseline	Blog	Index	Login	404	Baseline	Blog	Index	Login
Accessibility	2.29%	2.18%	2.3%	2.32%	2.37%	5.85%	5.72%	5.92%	6.05%	6.05%	268.94	209.8	289.09	280.79	245.24
Art & Design	4.62%	3.04%	4.83%	3.56%	4.67%	14.67%	14.93%	15.02%	14.94%	14.84%	275.82	241.05	278.04	271.94	263.11
Communication	2.6%	2.52%	2.63%	2.63%	2.63%	10.2%	10.6%	10.2%	10%	10.29%	337.35	392.35	371.15	375.74	370.66
Developer Tools	3.07%	2.95%	3.2%	3.23%	3.14%	7.87%	8.19%	8%	7.94%	8%	229.56	184.14	233.9	234.45	227.25
Education	3.07%	3.67%	3.24%	3.39%	3.28%	9.27%	9.81%	9.26%	9.6%	9.53%	420.23	401.12	394.65	462.3	416.25
Entertainment	6.28%	5.7%	5.86%	5.85%	5.83%	11.78%	11.91%	11.78%	11.91%	11.84%	230.65	189.94	226.94	233.66	223.36
Functionality and UI	1.94%	1.67%	1.97%	1.66%	1.93%	6.01%	6.15%	6.01%	6.01%	6.01%	194.25	164.14	203.02	193.26	189.3
Games	27.96%	31.16%	27.48%	28.36%	27.95%	62.73%	63.33%	62.99%	62.86%	62.43%	2047.23	2031.53	2027.97	2142.49	2105.75
Household	0.09%	0.08%	0.09%	0.09%	0.09%	5.26%	6.32%	5.26%	5.26%	5.26%	142.35	101.38	144.98	144.69	131.91
Just for fun	4.61%	4.37%	4.5%	4.73%	4.7%	12.25%	11.24%	12.25%	11.98%	12.25%	308.36	251.2	296.6	312.95	296.11
News & Weather	1.45%	1.28%	1.45%	1.41%	2.27%	3.99%	3.87%	3.86%	3.93%	3.93%	422.44	330.39	449.95	424.43	327.22
Privacy & Security	2.67%	2.48%	2.6%	2.6%	2.45%	13.78%	13.92%	13.94%	13.76%	14.06%	363.42	313.67	365.78	398.94	390.33
Shopping	15.26%	15.21%	14.79%	15.21%	14.83%	24.08%	24.05%	24.03%	23.83%	23.81%	643.9	615.82	623.94	648.8	626.21
Social Networking	3.8%	3.69%	3.85%	3.9%	3.73%	9.45%	9.45%	9.39%	9.2%	9.26%	672.21	791.21	580.82	538.64	635.5
Tools	8.17%	7.97%	8.43%	8.49%	8.32%	17.93%	17.73%	17.9%	18.01%	17.75%	458.14	415.2	460.4	495.87	482.17
Travel	0.67%	0.79%	0.68%	0.7%	0.67%	20.31%	24.22%	20.31%	20.31%	20.31%	277.77	260.08	262.11	271.26	262.38
Wellbeing	0.45%	0.39%	0.49%	0.51%	0.5%	7.1%	7.1%	7.12%	7.1%	7.1%	131.51	169.03	136.28	136.44	125.43
Workflow and planning	11%	10.82%	11.61%	11.35%	10.64%	22.85%	22.98%	22.82%	22.68%	22.53%	570.46	536.95	580.89	568.01	536.9

(b) Accessibility Results

Category	Altered Accessibility					Worse Accessibility					Better Accessibility				
	404	Baseline	Blog	Index	Login	404	Baseline	Blog	Index	Login	404	Baseline	Blog	Index	Login
Accessibility	3.33%	3.86%	3.59%	3.59%	3.52%	2.66%	2.13%	2.86%	2.86%	2.93%	1.66%	2.66%	1.86%	1.8%	1.73%
Art & Design	1.56%	1.65%	1.56%	1.56%	1.56%	1.22%	1.13%	1.22%	1.22%	1.22%	0.69%	0.95%	0.69%	0.78%	0.78%
Communication	5.25%	5.85%	5.66%	5.35%	5.85%	4.14%	4.34%	4.55%	4.24%	4.74%	4.55%	4.84%	4.55%	4.55%	4.54%
Developer Tools	1.8%	1.73%	1.87%	1.67%	1.93%	1.13%	1.07%	1.27%	1.07%	1.47%	1.2%	1.2%	1.13%	1.13%	1.07%
Education	4.04%	4.94%	4.45%	4.38%	4.45%	3.07%	3.2%	3.41%	3.27%	3.48%	2.37%	3.13%	2.44%	2.51%	2.44%
Entertainment	0.87%	1.13%	1%	0.86%	0.93%	0.8%	1%	0.93%	0.8%	0.86%	0.33%	0.4%	0.33%	0.33%	0.27%
Functionality and UI	1.47%	1.54%	1.54%	1.47%	1.54%	1.2%	1.07%	1.34%	1.27%	1.34%	0.6%	0.8%	0.53%	0.53%	0.53%
Games	0.33%	0.4%	0.33%	0.33%	0.33%	0.27%	0.27%	0.27%	0.27%	0.27%	0.13%	0.2%	0.13%	0.13%	0.13%
Household	2.11%	2.11%	2.11%	2.11%	2.11%	2.11%	2.11%	2.11%	2.11%	2.11%	1.05%	1.05%	1.05%	1.05%	1.05%
Just for fun	2.33%	2.46%	2.4%	2.26%	2.33%	1.86%	1.66%	1.93%	1.86%	1.86%	1.07%	1.4%	1.07%	1%	1%
News & Weather	1.13%	1.27%	1.13%	1.07%	1.26%	1%	1.07%	1%	0.93%	1.13%	0.67%	0.73%	0.6%	0.6%	0.6%
Privacy & Security	2.56%	4.16%	3.53%	2.56%	3.83%	1.44%	2.4%	2.08%	1.44%	2.56%	1.76%	3.36%	2.4%	1.76%	2.88%
Shopping	5%	5.41%	5.21%	5.01%	4.88%	4.07%	4.01%	4.27%	4.27%	4.08%	2.33%	2.81%	2.4%	2.27%	2.07%
Social Networking	2.33%	2.26%	2.27%	2.33%	2.4%	1.73%	1.66%	1.67%	1.73%	1.73%	1.4%	1.46%	1.4%	1.33%	1.4%
Tools	4.2%	4.4%	4.21%	4.14%	4.34%	3.27%	3%	3.41%	3.34%	3.6%	2.4%	2.8%	2.27%	2.27%	2.2%
Travel	0.78%	1.56%	0.78%	0.78%	0.78%	0.78%	1.56%	0.78%	0.78%	0.78%	0.78%	0.78%	0.78%	0.78%	0.78%
Wellbeing	1.14%	2.27%	1.14%	1.42%	1.14%	0.85%	1.42%	1.14%	1.14%	1.14%	0.85%	1.42%	0.85%	0.85%	0.85%
Workflow and planning	4.53%	5.34%	4.67%	4.47%	4.53%	3.6%	3.41%	3.74%	3.47%	3.67%	2.93%	4.01%	2.94%	2.87%	2.87%

Table 18: Overview of Accessibility Results Across Honeypages

	Extensions	Permissions					Manifest		API Calls					Ratings					No. Users				
		Mean	Median	Mode	Min	Max	V2	V3	Mean	Median	Mode	Min	Max	Mean	Median	Mode	Min	Max	Mean	Median	Mode	Min	Max
WordPress Index																							
Overall	100%	1.97	1	0	0	18	27.31%	72.61%	508	49	38	0	440K	4.04	4.6	5	0	5	57K	285	1000	0	206M
Not Altered	97.34%	1.93	1	0	0	18	27.46%	72.46%	477	48	38	0	440K	4.04	4.6	5	0	5	56K	278	1000	0	206M
Altered	2.66%	3.65	3	2	0	16	21.91%	78.09%	1623	468	615	0	77K	4.18	4.7	5	0	5	93K	538	1000	0	6M
Improved	1.5%	3.75	3	3	0	16	21.56%	78.44%	1947	545	615	0	77K	4.09	4.6	5	0	5	110K	430	0	0	6M
Worse	2.1%	3.68	3	2	0	16	21.48%	78.52%	1427	512	615	0	61K	4.17	4.7	5	0	5	63K	476	1000	0	5M
WordPress Login																							
Overall	100%	1.97	1	0	0	18	27.3%	72.62%	490	39	28	0	323K	4.04	4.6	5	0	5	57K	285	1000	0	206M
Not Altered	97.22%	1.93	1	0	0	18	27.41%	72.51%	465	38	28	0	323K	4.04	4.6	5	0	5	56K	278	1000	0	206M
Altered	2.78%	3.57	3	2	0	16	23.52%	76.48%	1378	405	605	0	76K	4.18	4.6	5	0	5	94K	500	1000	0	6M
Improved	1.5%	3.75	3	2	0	16	23.12%	76.88%	1686	495	605	0	76K	4.08	4.6	5	0	5	114K	430	0	0	6M
Worse	2.25%	3.6	3	2	0	16	22.59%	77.41%	1457	436	605	0	76K	4.17	4.7	5	0	5	64K	452	1000	0	5M
WordPress Blog																							
Overall	100%	1.97	1	0	0	18	27.3%	72.62%	495	49	38	0	485K	4.04	4.6	5	0	5	57K	285	1000	0	206M
Not Altered	97.23%	1.93	1	0	0	18	27.48%	72.44%	467	48	38	0	485K	4.04	4.6	5	0	5	56K	278	1000	0	206M
Altered	2.77%	3.62	3	2	0	16	21.02%	78.98%	1466	445	615	0	77K	4.2	4.7	5	0	5	93K	518	1000	0	6M
Improved	1.54%	3.76	3	3	0	16	21.04%	78.96%	1693	506	615	0	77K	4.1	4.6	5	0	5	114K	454	0	0	6M
Worse	2.2%	3.64	3	2	0	16	20.3%	79.7%	1433	505	615	0	41K	4.19	4.7	5	0	5	61K	452	1000	0	5M
WordPress 404																							
Overall	100%	1.97	1	0	0	18	27.31%	72.61%	500	49	38	0	443K	4.04	4.6	5	0	5	57K	286	1000	0	206M
Not Altered	97.36%	1.93	1	0	0	18	27.45%	72.47%	476	48	38	0	443K	4.04	4.6	5	0	5	56K	278	1000	0	206M
Altered	2.64%	3.68	3	2	0	16	22.1%	77.9%	1393	448	615	0	77K	4.18	4.6	5	0	5	97K	576	1000	0	6M
Improved	1.52%	3.76	3	2	0	16	21.6%	78.4%	1558	490	615	0	77K	4.1	4.6	5	0	5	115K	430	0	0	6M
Worse	2.06%	3.71	3	2	0	16	21.69%	78.31%	1404	508	615	0	49K	4.16	4.7	5	0	5	64K	512	1000	0	5M

5 Accessibility Results

Table 17 shows a summary of the accessibility results for all of our HoneyPages.

Table 17: Summary of Accessibility Results (I = Increases, D = Decreases)

HoneyPage	Lighthouse		WAVE											
			Errors		Contrast Errors		Alerts		Features		Structural Elements		ARIA	
	I	D	I	D	I	D	I	D	I	D	I	D	I	D
	Baseline	145	235	242	5	37	30	114	0	77	6	251	5	77
Wordpress Index	0	310	233	0	31	0	115	6	70	7	248	4	74	5
Wordpress Login	0	334	235	0	43	0	119	0	71	7	245	5	75	5
Wordpress Blog	10	326	239	0	37	0	120	10	70	12	246	6	76	7
Wordpress 404	12	291	237	0	32	0	117	6	69	13	245	7	74	8

Table 19: Top API Calls for Extensions that Alter Google Lighthouse Scores on the Baseline HoneyPage

(a) Worse Lighthouse Score		(b) Same Lighthouse Score	
API Name	Percentage	API Name	Percentage
CSSStyleDeclaration	17.33	HTMLInputElement	14.88
Global	12.03	HTMLDivElement	11.93
HTMLDivElement	10.91	Object	11.12
HTMLSpanElement	6.21	HTMLAnchorElement	7.59
HTMLDocument	4.73	CSSStyleDeclaration	4.53
HTMLStyleElement	4.21	HTMLDocument	4.39
DOMTokenList	3.92	IDBRequest	3.52
Window	2.49	Window	2.81
Object	2.32	HTMLImageElement	2.80
HTMLElement	1.99	HTMLScriptElement	2.14
(c) Better Lighthouse Score			
API Name		Percentage	
CSSStyleDeclaration		15.49	
HTMLDivElement		14.64	
HTMLDocument		5.88	
Object		3.89	
HTMLAnchorElement		3.87	
Window		2.93	
HTMLSpanElement		2.37	
DOMTokenList		2.25	
HTMLInputElement		2.14	
HTMLStyleElement		2.09	

Table 20: Mean Lighthouse Score per Permission for the Baseline Honeypage (Top and Bottom 5 Permissions Shown)

Permission	Lighthouse (%)	WAVE					
		Errors	Contrast Error	Alerts	Features	Structural Elements	ARIA
fontSettings	86.54	1.15	2.77	0.00	3.00	6.00	4.00
desktopCapture	86.10	1.41	3.00	0.17	3.12	6.10	4.43
debugger	86.05	1.08	3.00	0.14	3.00	6.02	4.09
downloads.shelf	86.04	1.00	3.30	0.00	3.00	6.19	4.00
tabGroups	86.04	1.08	3.00	0.00	3.00	6.06	4.00
privacy	86.01	1.02	3.00	0.02	3.01	6.03	4.00
proxy	86.01	1.00	3.00	0.00	3.02	6.00	4.00
identity.email	86.00	1.11	3.03	0.05	3.03	6.03	4.21
topSites	86.00	1.02	3.00	0.00	3.00	6.01	4.00
Without Extension	86.00	1.00	3.00	0.00	3.00	6.00	4.00
system.cpu	86.00	1.00	3.00	0.00	3.00	6.00	4.00
ttsEngine	86.00	1.00	3.00	0.00	3.00	6.20	4.00
system.memory	86.00	1.00	3.00	0.00	3.00	6.00	4.00
power	86.00	1.00	3.00	0.00	3.00	6.00	4.00
downloads.open	86.00	1.00	3.42	0.00	3.00	6.00	4.00
accessibilityFeatures.read	86.00	1.00	3.00	0.00	3.00	6.00	4.00
fileBrowserHandler	86.00	1.00	3.00	0.00	3.00	6.00	4.00
webRequestAuthProvider	86.00	1.01	3.00	0.02	3.01	6.00	4.00
system.storage	86.00	1.00	3.00	0.00	3.00	6.00	4.00
declarativeNetRequestWithHostAccess	85.99	1.03	3.01	0.06	3.18	6.02	4.04
None	85.99	1.00	3.00	0.01	3.00	6.00	4.00
nativeMessaging	85.99	0.99	3.02	0.03	3.03	6.00	4.06
geolocation	85.98	1.02	2.98	0.00	3.01	6.03	4.00
idle	85.98	1.06	3.00	0.02	3.00	6.06	4.00
downloads	85.97	1.06	3.01	0.03	3.01	6.05	4.06
declarativeContent	85.96	1.00	3.00	0.00	3.00	6.00	4.00
favicon	85.96	1.06	3.00	0.00	3.00	6.04	4.00
tabCapture	85.96	1.08	3.00	0.11	3.00	6.08	4.08
system.display	85.96	1.03	3.00	0.05	3.00	6.03	4.01
identity	85.93	1.11	3.01	0.03	3.02	6.05	4.08
webRequest	85.93	1.04	2.99	0.05	3.02	6.00	4.04
storage	85.92	1.07	3.00	0.03	3.02	6.03	4.04
chrome://favicon/	85.92	1.00	3.00	0.00	3.00	6.02	4.00
tabs	85.91	1.10	3.00	0.05	3.03	6.05	4.04
activeTab	85.91	1.07	3.00	0.05	3.02	6.04	4.02
alarms	85.91	1.06	3.00	0.01	3.03	6.02	4.01
scripting	85.91	1.08	3.00	0.04	3.03	6.04	4.04
webNavigation	85.90	1.07	2.99	0.03	3.03	6.04	4.02
webRequestBlocking	85.90	1.03	2.98	0.02	3.00	6.01	4.03
unlimitedStorage	85.89	1.06	3.00	0.00	3.01	6.03	4.04
contentSettings	85.89	1.00	3.00	0.00	3.00	6.02	4.00
bookmarks	85.89	1.34	2.98	0.13	3.03	6.16	4.00
browsingData	85.88	1.25	2.95	0.01	3.08	6.11	4.00
search	85.88	1.62	3.00	0.04	3.35	6.27	4.00
sessions	85.88	1.04	3.00	0.00	3.00	6.00	4.00
declarativeNetRequest	85.87	1.08	3.00	0.00	3.00	6.00	3.99
gcm	85.87	1.20	3.00	0.01	3.03	6.03	4.00
clipboardRead	85.87	1.04	3.01	0.00	3.01	6.02	4.00
notifications	85.85	1.10	3.02	0.01	3.01	6.04	4.02
declarativeNetRequestFeedback	85.85	1.17	2.97	0.00	3.03	6.03	4.00
contextMenus	85.85	1.16	3.01	0.06	3.04	6.07	4.09
background	85.84	1.07	2.98	0.00	3.01	6.02	4.00
clipboardWrite	85.83	1.09	3.00	0.08	3.06	6.05	4.01
cookies	85.82	1.14	3.00	0.04	3.00	6.08	4.10
history	85.81	1.21	3.03	0.18	3.15	6.16	4.00
tts	85.81	1.04	2.96	0.03	3.00	6.07	4.15
windows	85.80	1.00	3.00	0.00	3.00	6.05	4.00
offscreen	85.76	1.11	3.01	0.06	3.00	6.05	4.07
sidePanel	85.75	1.09	3.00	0.01	3.01	6.04	4.05
commands	85.71	1.31	3.00	0.02	3.00	6.10	4.31
management	85.45	1.07	3.02	0.02	3.04	6.13	4.00
pageCapture	85.00	1.14	3.00	0.00	3.00	6.07	4.00

Table 21: Extensions which Altered an Accessibility Score for the Security & Privacy Category, sorted by Lighthouse Score

Permission	Lighthouse (%)	WAVE					
		Errors	Contrast Error	Alerts	Features	Structural Elements	ARIA
Tibo	91	1	4	0	4	6	9
Sentinel Secure	89	1	3	1	3	7	4
Swift Security Extension	89	1	8	2	4	6	4
AdBlock - Ads and Youtube	88	1	3	0	3	6	4
nlogin.me	88	1	3	0	3	6	4
LockSelf	87	1	3	0	3	8	4
WOT Website Security & Safety Checker	86	1	3	0	6	6	4
DualSafe Password Manager & Digital Vault	86	1	3	0	3	7	4
StrongPass	86	2	3	2	4	6	4
SaudiCircuit	86	1	1	1	5	6	4
MailMask	86	1	3	1	3	6	4
HTTPS Everywhere	86	1	4	0	3	6	4
Baseline without Extension	86	1	3	0	3	6	4
Cyber Control Your Data & Earn Rewards	86	2	3	0	3	6	4
Anti Phishing - Safer Browsing for Chrome	86	1	3	0	4	7	4
Vpn Switcher 5 addresses free	86	1	3	0	4	6	4
IronVest	86	1	3	0	3	7	4
Snap VPN	86	1	3	0	4	6	4
Safum free VPN	86	1	3	0	4	6	4
Blaze VPN	86	1	3	0	4	6	4
EviCypher Webmail & Cloud	85	0	3	0	4	6	7
PassCypher HSM PGP	85	0	5	0	4	6	7
Canary	85	1	3	0	3	7	4
Buttercup	80	1	3	0	3	6	6
Open incognito tab	79	2	3	0	3	6	4
Keptler Email Masking	79	2	3	0	3	6	4
Trend Micro ID Protection	74	2	3	0	3	7	4

Table 22: Extensions which Altered the Lighthouse Accessibility Score for the Accessibility Category

Permission	Lighthouse (%)	WAVE					
		Errors	Contrast Error	Alerts	Features	Structural Elements	ARIA
Dark Reader	93	1	0	0	3	6	4
Sauron - Dark mode for all websites	93	1	0	0	3	6	4
Dark Theme - Dark Reader for Web	93	1	0	0	3	6	4
Illumify	93	1	0	0	3	6	4
Low Vision Webpage Layout Modifier	93	1	0	0	3	6	4
Dark Mode Master	93	1	0	0	3	6	4
Babu Background RGB Color	93	1	0	0	3	6	4
Super Dark Mode	93	1	0	0	3	6	4
Special Dark	93	1	0	0	3	6	4
Dark mode night reader	93	1	0	0	3	6	4
Dark Mode - Night Eye	93	1	0	0	3	6	4
dark reader - dark mode for Chrome	93	1	0	0	3	6	4
Dark Mode - Dark Reader for Chrome	93	1	0	0	3	6	4
Midnight Lizard	93	1	0	0	3	6	4
Flexpad	88	1	3	0	3	6	4
Style Tab	88	1	0	0	3	8	4
QuartTools	88	23	3	27	27	11	104
Pinyin Generator	88	1	3	2	3	10	4
Easy Reader	87	1	3	0	3	6	5
Web Color Filter	87	1	3	0	3	6	5
Password Show Hide	86	1	3	1	3	6	4
Video Speed Controls	86	3	3	0	3	6	4
Color Vision Deficiency Assistant	86	1	3	1	3	6	4
YouWatch	86	1	3	1	3	6	4
Selection Translator	86	1	3	1	3	6	4
3asyR	86	5	3	6	4	6	10
QR Code Safety	86	2	3	0	3	6	4
AutoScrolling	86	13	3	2	3	12	6
Pixie	86	2	3	0	3	7	4
iTour Video Translation	86	5	3	0	3	6	12
The Rayo Assistant	86	7	3	12	10	9	4
DragonLink	86	2	3	0	3	6	4
Screen Recorder	86	1	3	1	3	6	4
Post! (Postila)	86	1	3	1	3	6	4
Cool Translator	86	1	3	0	3	7	4
AnyConnect Check	86	1	3	0	3	7	4
VidHelper - Video Download Helper	86	1	3	0	3	7	4
Sky Reader	86	1	0	0	3	6	4
Video Image Control (new gamma)	86	1	3	0	3	7	4
QReaderX - QR Code Reader	86	1	3	0	3	6	5
carVertical Vehicle History & VIN Lookup	86	1	3	0	3	7	4
Greeked and blurred	86	1	3	0	3	7	4
Baseline without Extension	86	1	3	0	3	6	4
Dyslexic Browser	86	5	3	0	3	6	4
ReadText (Text to Speech) by MagicLabs	86	1	3	2	3	6	4
Dark Night Mode	86	1	5	0	3	6	4
Proper Menubar for Google Chrome	84	1	3	0	3	6	5
NoPhish	82	2	3	0	3	8	4
Pericles Text to Speech Screen Reader	81	1	3	0	3	7	4
Collie Chrome Extension	81	5	3	0	8	6	4
Fluid Bookmarks	81	1	3	0	3	7	4
Bookpin	81	1	3	0	3	7	4
YouTube Dark Mode	79	2	3	0	3	6	4
Readable - dyslexia helper	79	5	8	1	3	7	4
Sloth	79	3	3	0	3	6	4
Volume Booster	79	2	3	1	3	6	4
Scroll buttons	79	3	3	0	3	6	4
Volume Controller	79	2	3	1	3	6	4
Click-free Browsing	79	16	3	0	3	6	4

Table 23: Summary of WAVE Errors Increased for Extensions which Altered Accessibility for the Accessibility Category

Wave Error	Number of Extensions
Missing alternative text	33
Missing form label	24
Empty form label	10
Broken ARIA reference	9
Empty button	6
Multiple form labels	5
Linked image missing alternative text	1